

Outputs

main.py Terminal output:

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Original APOD Image Information:

Awkward and angular looking, Apollo 17's lunar module Challenger was designed for flight in the near vacuum of space. Digitally enhanced and reprocessed, this picture taken from Apollo 17's command module America shows Challenger's ascent stage in lunar orbit. Small reaction control thrusters are at the sides of the moonship with the bell of the ascent rocket engine underneath. The hatch allowing access to the lunar surface is seen at the front, with a round radar antenna at the top. Mission commander Gene Cernan is clearly visible through the triangular window. This spaceship performed gracefully, landing on the Moon and returning the Apollo astronauts to the orbiting command module in December of 1972. So where is Challenger now? Its descent stage remains at the Apollo 17 landing site in the Taurus-Littrow valley. The ascent stage pictured was intentionally crashed nearby after being jettisoned from the command module prior to the astronauts' return to planet Earth.

RAG Generated Description:

The photo shows the Lunar module Challenger, which was part of the Apollo 17 mission, in its ascent stage in lunar orbit. The image was taken from the command module America, where astronaut Gene Cernan is clearly visible through a triangular window.

The Lunar Module, also known as the Apollo Lunar Module (LM), was a crucial component of the Apollo program. It was designed to fly between lunar orbit and the Moon's surface, carrying two astronauts on a historic mission to explore the lunar surface. The LM was structured in two stages: an ascent stage that flew back to the command module after landing, and a descent stage that remained on the lunar surface.

The ascent stage had small reaction control thrusters (RCS) attached to its sides, which were used for attitude control and navigation. Below the RCS, there was a bell-shaped rocket engine that provided thrust during takeoff and ascent. The hatch at the front of the LM allowed astronauts like Gene Cernan to access the lunar surface.

Reaction Control Systems (RCS) are an essential component of space exploration. They use thrusters to provide attitude control and translation in any desired direction or combination of directions. In this image, the RCS can be seen as small modules on the sides of the ascent stage.

The Lunar Module was developed by Grumman Aerospace Engineering Corporation, which faced significant challenges during its development. Despite these obstacles, the LM became one of the most reliable components of the Apollo-Saturn space vehicle.

Ten Lunar Modules were launched into space, with six successfully landing on the Moon's surface between 1969 and 1972. The first two flown were test missions in low Earth orbit: Apollo 5 without a crew and Apollo 9 with a crew. A third test flight in low lunar orbit was conducted by Apollo 10, which served as a dress rehearsal for the first landing on Apollo 11.

The six landed descent stages remain at their respective landing sites, while the corresponding ascent stages were either discarded in heliocentric orbits or destroyed during controlled re-entry into Earth's atmosphere. One ascent stage (Apollo 18's Snoopy) was intentionally jettisoned from the command module before returning to Earth.

Taurus-Littrow Valley, where Apollo 17 landed in December 1972, is a unique lunar feature located on the Moon's near side. The valley is situated at coordinates 20.0°N 31.0°E and features an impact basin formed about 3.8 billion years ago. The region includes Littrow crater to the south, ring mountains surrounding it, and Mare Serenitatis to the east.

The name Taurus-Littrow was coined by the Apollo 17 crew, who selected this site for its unique geological features. Data collected during the mission indicate that the valley is composed primarily of feldspar-rich breccia in the surrounding massifs and basalt underlying the valley floor. The image provides a glimpse into the rich geological history of the Moon's surface, which was further explored by the Apollo 17 astronauts.

The Lunar Module Challenger played a pivotal role in human exploration of the Moon, serving as the vehicle that landed on the lunar surface during the Apollo 17 mission. Its development and deployment paved the way for future space missions, demonstrating the ingenuity and determination of NASA engineers and astronauts.
Image successfully saved as nasa_apod_image.jpg
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Result.txt:

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Original APOD Image Information:

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Additional Search Terms:
• 'Apollo'
• 'Challenger'
• 'Lunar Module'
• 'Reaction Control Thrusters'
• 'Ascent Stage'
• 'Descent Stage'
• 'Mission Commander'
• 'Gene Cernan'
• 'Taurus-Littrow Valley'

Searching Wikipedia for 'Apollo'...
Summary of Apollo: In ancient Greek religion and mythology, Apollo is one of the Olympian deities. His numerous functions include healing, prophecy, music, poetry, and archery. He is the son of Zeus and Leto, and the twin brother of Artemis, goddess of the hunt. He is considered to be the most beautiful god and is represented as the ideal of the kouros (ephebe, or a beardless, athletic youth). In the 5th century BC, his worship was imported to Rome. As the patron deity of Delphi (Apollo Pythios), Apollo is an oracular god-the prophetic deity of the Delphic Oracle and the deity of ritual purification. His oracles were often consulted for guidance in various matters. He was in general seen as the god who affords help and wards off evil, and is referred to as Alexiakaos, the "avert-er of evil". Medicine and healing are associated with Apollo, whether through the god himself or mediated through his son Asclepius. Apollo delivered people from epidemics, yet he is also a god who could bring ill health and deadly plague with his arrows. The invention of archery itself is credited to Apollo and his sister Artemis. Apollo is usually described as carrying a silver or golden bow and a quiver of arrows. As the god of music, Apollo presides over all music, songs, dance, and poetry. He is the inventor of string-music and the frequent companion of the Muses, functioning as their chorus leader in celebrations. The lyre is a common attribute of Apollo. Protection of the young is one of the best attested facets of his panhellenic cult persona. As a kourotrophos, Apollo is concerned with the health and education of children, and he presided over their passage into adulthood. Long hair, which was the prerogative of boys, was cut at the coming of age (ephebeia) and dedicated to Apollo. The god himself is depicted with long, uncut hair to symbolise his eternal youth. Apollo is an important pastoral deity, and he was the patron of herdsmen and shepherds. Protection of herds, flocks and crops from diseases, pests and predators were his primary rustic duties. On the other hand, Apollo also encouraged the founding of new towns and the establishment of civil constitutions, is associated with dominion over colonists, and was the giver of laws. His oracles were often consulted before setting laws in a city. Apollo Agnæus was the protector of the streets, public places and home entrances. From the 5th century BCE, Apollo was often identified with Helios, the personification of the Sun. Although Latin theological works from at least 1st century BCE identified Apollo with Sol, there was no conflation between the two among the classical Latin poets until 1st century CE. Apollo is known in Greek-influenced Etruscan mythology as Apulu.

Searching Wikipedia for 'Challenger'...
An issue occurred retrieving information for 'Challenger'.

Searching Wikipedia for 'Lunar Module'...
Summary of Apollo Lunar Module: The Apollo Lunar Module (LM ), originally designated the Lunar Excursion Module (LEM), was the lunar lander spacecraft that was flown between lunar orbit and the Moon's surface during the United States' Apollo program. It was the first crewed spacecraft to operate exclusively in space, and remains the only crewed vehicle to land anywhere beyond Earth. Structurally and aerodynamically incapable of flight through Earth's atmosphere, the two-stage Lunar Module was ferried to lunar orbit attached to the Apollo command and service module (CSM), about twice its mass. Its crew of two flew the Lunar Module from lunar orbit to the Moon's surface. During takeoff, the spent descent stage was used as a launch pad for the ascent stage which then flew back to the command module, after which it was also discarded. Overseen by Grumman, the LM's development was plagued with problems that delayed its first uncrewed flight by about ten months and its first crewed flight by about three months. Regardless, the LM became the most reliable component of the Apollo-Saturn space vehicle. The total cost of the LM for development and the units produced was $21.65 billion in 2016 dollars, adjusting from a nominal total of $2.29 billion using the NASA New Start Inflation Indices. Ten Lunar Modules were launched into space. Of these, six were landed by humans on the Moon from 1969 to 1972. The first two flown were tests in low Earth orbit: Apollo 5, without a crew; and Apollo 9 with a crew. A third test flight in low lunar orbit was Apollo 10, a dress rehearsal for the first landing, conducted on Apollo 11. The Apollo 13 Lunar Module functioned as a lifeboat to provide life support and propulsion to keep the crew alive for the trip home, when their CSM was disabled by an oxygen tank explosion en route to the Moon. The six landed descent stages remain at their landing sites; their corresponding ascent stages crashed into the Moon following use. One ascent stage (Apollo 18's Snoopy) was discarded in a heliocentric orbit after its descent stage was discarded in lunar orbit. The other three LMs were destroyed during controlled re-entry in the Earth's atmosphere: the four stages of Apollo 5 and Apollo 9 each re-entered separately, while Apollo 13's Aquarius re-entered as a unit.

Searching Wikipedia for 'Reaction Control Thrusters'...
Summary of Reaction control system: A reaction control system (RCS) is a spacecraft system that uses thrusters to provide attitude control and translation. Alternatively, reaction wheels can be used for attitude control, rather than RCS. Use of diverted engine thrust to provide stable attitude control of a short-or-vertical takeoff and landing aircraft below conventional winged flight speeds, such as with the Harrier "jump jet", may also be referred to as a reaction control system. Reaction control systems are capable of providing small amounts of thrust in any desired direction or combination of directions. An RCS is also capable of providing torque to allow control of rotation (roll, pitch, and yaw). Reaction control systems often use combinations of large and small (vernier) thrusters, to allow different levels of response.

Searching Wikipedia for 'Ascent Stage'...
Summary of Apollo 18: Apollo 1, initially designated AS-204, was planned to be the first crewed mission of the Apollo program, the American undertaking to land the first man on the Moon. It was planned to launch on February 21, 1967, as the first low Earth orbital test of the Apollo command and service module. However, the mission never flew; a cabin fire during a launch rehearsal test at Cape Kennedy Air Force Station Launch Complex 34 on January 27, 1967 killed all three crew members-Command Pilot Gus Grissom, Senior Pilot Ed White, and Pilot Roger B. Chaffee-and destroyed the command module (CM). The name Apollo 1, chosen by the crew, was made official by NASA in their honor after the fire.
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The Lunar Module Challenger played a pivotal role in human exploration of the Moon, serving as the vehicle that landed on the lunar surface during the Apollo 17 mission. Its development and deployment paved the way for future space missions, demonstrating the ingenuity and determination of NASA engineers and astronauts.

Link to this photo:
<https://apod.nasa.gov/apod/image/2412/AS17-149-22859-2v25n1mk.jpg>

Immediately after the fire, NASA convened an Accident Review Board to determine the cause of the fire, and both chambers of the United States Congress conducted their own committee inquiries to oversee NASA's investigation. The ignition source of the fire was determined to be electrical, and the fire spread rapidly due to combustible nylon material and the high-pressure pure oxygen cabin atmosphere. Rescue was prevented by the plug door hatch, which could not be opened against the internal pressure of the cabin. Because the rocket was unfueled, the test had not been considered hazardous, and emergency preparedness for it was poor. During the congressional investigation, Senator Walter Mondale publicly revealed a NASA internal document citing problems with prime Apollo contractor North American Aviation, which became known as the Phillips Report. This disclosure embarrassed NASA Administrator James E. Webb, who was unaware of the document's existence, and attracted controversy to the Apollo program. Despite congressional displeasure at NASA's lack of openness, both congressional committees ruled that the issues raised in the report had no bearing on the accident. Crewed Apollo flights were suspended for twenty months while the command module's hazards were addressed. However, the development and uncrewed testing of the Lunar module (LM) and Saturn V rocket continued. The Saturn IB launch vehicle for Apollo 1, AS-204, was used for the first LM test flight, Apollo 5. The first successful crewed Apollo mission was flown by Apollo 1's backup crew on Apollo 7 in October 1968.

Searching Wikipedia for 'Descent Stage'...

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Searching Wikipedia for 'Mission Commander'...

An issue occurred retrieving information for 'Mission Commander'.

Searching Wikipedia for 'Gene Cernan'...

An issue occurred retrieving information for 'Gene Cernan'.

Searching Wikipedia for 'Taurus-Littrow Valley'...

Summary of Taurus-Littrow: Taurus-Littrow is a lunar valley located on the near side at the coordinates 20.0°N 31.0°E / 20.0; 31.0. It served as the landing site for the American Apollo 17 mission in December 1972, the last crewed mission to the Moon. The valley is located on the southeastern edge of Mare Serenitatis along a ring of mountains formed between 3.8 and 3.9 billion years ago when a large object impacted the Moon, forming the Serenitatis basin and pushing rock outward and upward. Taurus-Littrow is located in the Taurus mountain range and south of Littrow crater, features from which the valley received its name. The valley's name, coined by the Apollo 17 crew, was approved by the International Astronomical Union in 1973. Data collected during Apollo 17 indicate that the valley is composed primarily of feldspar-rich breccia in the large massifs surrounding the valley and basalt underlying the valley floor, covered by an unconsolidated layer of mixed material formed by various geologic events. Taurus-Littrow was selected as the Apollo 17 landing site with the objectives of sampling highland material and young volcanic material at the same location.

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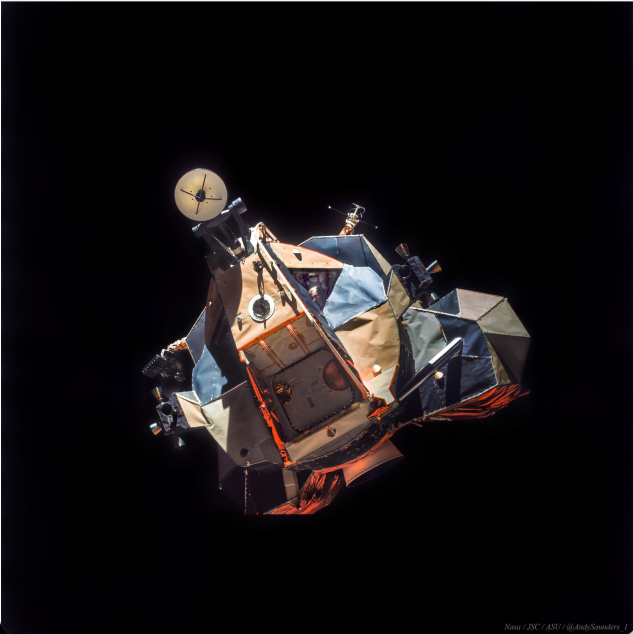
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Photo of the day:



Nasa / JSC / AS17-149-22859-2v25n1mk.jpg